Tanmay Kumar

Indian Institute of Technology (IIT) Mandi · B.Tech in Civil Engineering (3rd Year)
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Education

Indian Institute of Technology (IIT) Mandi — Mandi, Himachal Pradesh, India
Bachelor of Technology in Civil Engineering (Current CGPA / Standing: 3rd Year) — *2022 – 2026 (Expected)*

- **Focus Areas:** Computational Mechanics, Scientific Machine Learning (SciML), Stochastic Reliability Analysis, Finite Element Methods (FEM), Fluid-Structure Multiphysics, and Numerical Optimization.

Technical Skills

Languages & Core: Python (NumPy, SciPy, PyTorch, Scikit-Learn), C/C++, TypeScript, JavaScript, SQL, Bash, LaTeX
Scientific Computing: Physics-Informed Neural Networks (PINNs, XPINNs), Fourier Neural Operators (FNO), DeepONet, Finite Element Analysis (FEA/FEM), Monte Carlo, Subset Simulation, FORM/SORM, Proper Orthogonal Decomposition (ROM)
Engineering Standards: ISO 2394 (Structural Reliability), DNV-RP-C210 (Offshore Fatigue & Fracture), Eurocode 3 (EN 1993-1-2 Structural Fire Design), ISO 13571 (Life Safety / Tenability)
Web & Systems Engineering: FastAPI, React 18, Vite, Three.js (WebGL 3D Rendering), Recharts, Docker, Git, CI/CD, KaTeX

Featured Engineering & Scientific Projects

CherenkovLens | *Python, Scikit-Learn, SciPy, Astrophysics, Hillas Eigensolver, Vercel* — Live Demo | GitHub

- Engineered an end-to-end simulation, topological image cleaning, and machine learning pipeline for **Imaging Atmospheric Cherenkov Telescopes (IACTs)**, modeled on the 21m **MACE Observatory** in Hanle, Ladakh (4,270 m a.s.l.).
- Simulated relativistic Extensive Air Shower (EAS) microphysics via the **Frank-Tamm equation** and barometric refractive index profiles across a 34×34 camera pixel array (1,156 PMT channels), separating electromagnetic cascades from hadronic proton backgrounds.
- Implemented dual-threshold topological tail-cuts cleaning and an analytical **Hillas eigensolver** computing statistical image moments (Length, Width, Dist, Alpha, Azwidth, Size, Concentration).
- Trained Random Forest and SVM classifiers achieving high gamma/hadron discrimination, deployed via an interactive client-side web application featuring real-time parameter visualization and KaTeX mathematical formulations.

Reliability-Lab | *Python, FastAPI, React, Vite, Stochastic Reliability, Copulas, ISO 2394, DNV-RP-C210* Live Demo | GitHub

- Developed an enterprise-grade structural reliability and uncertainty quantification platform compliant with **ISO 2394** and **DNV-RP-C210** probabilistic safety assessment standards.
- Formulated First- and Second-Order Reliability Methods (**FORM/SORM**) with Hasofer-Lind optimization and Rackwitz-Fiessler normal-tail transformations for nonlinear limit state functions.
- Deployed advanced rare-event simulation algorithms (**Subset Simulation, Importance Sampling, Directional Simulation**) estimating extreme failure probabilities ($P_f \in [10^{-9}, 10^{-2}]$) with low coefficient of variation.
- Modeled heavy-tailed multivariate dependencies using Archimedean copulas (Clayton, Gumbel, Frank) and Gaussian/Student-t copulas; executed **Sobol variance-based global sensitivity analysis** and minimal cut-set system reliability.
- Architected a responsive full-stack platform (FastAPI + React/Vite) with real-time SVG limit-state contour plotting and automated numerical provenance verification.

firetwin | *Python 3.12, PyTorch, FastAPI, React 18, TypeScript, FNO, LES CFD, Eurocode 3* — Live Demo | GitHub

- Created a coupled multiphysics digital twin unifying Incompressible Boussinesq Navier-Stokes hydrodynamics, Smagorinsky-Lilly **Large Eddy Simulation (LES)**, and P1 radiation transport for compartment fire hazards.
- Trained a 2D **Fourier Neural Operator (FNO)** surrogate enabling continuous velocity and temperature field inference in < 5 **ms** ($> 100\times$ speedup over numerical CFD solvers) for real-time safety exploration.
- Coupled compartment gas temperatures to structural steel members via **Eurocode 3 (EN 1993-1-2)** lumped-capacitance formulations, evaluating thermal yield degradation $k_y(T)$ and elastic modulus reduction $k_E(T)$.
- Integrated **Helbing Social Force** pedestrian evacuation dynamics with ISO 13571 Fractional Effective Dose (FED) toxic incapacitation models to quantify Available Safe Egress Time (ASET) vs. Required Safe Egress Time (RSET).
- Engineered an automated serverless engine producing publication-grade 3-page compliance PDF reports complete with 6 embedded telemetry charts and Eurocode fire resistance matrices.

Neural PDE Bench | *PyTorch, Python, FastAPI, PINNs, FNO, DeepONet, Bayesian UQ, Diffusion Models* Live Demo | GitHu

- Built a comprehensive scientific machine learning platform integrating **Physics-Informed Neural Networks (PINNs)** and operator learning for forward and inverse partial differential equations.
- Implemented Residual-Based Adaptive Refinement (**RAR**) and multi-subdomain **XPINN decomposition** with interface flux-matching penalties to resolve stiff gradient fronts and multiscale dynamics.

- Benchmarked operator learning architectures including **Fourier Neural Operators (FNO-1D/2D)**, **DeepONet** (branch-trunk networks), and Geometry Graph Neural Operators (GNO) across continuum mechanics benchmarks.
- Conducted Bayesian Uncertainty Quantification via **Hamiltonian Monte Carlo (HMC)** and Variational MC-Dropout; deployed score-based diffusion generative models for inverse spatial parameter reconstruction $E(x)$.
- Constructed an interactive React/TypeScript analytical suite with real-time Recharts dashboards and a containerized FastAPI backend with automated pytest CI pipelines.

Nonlinear FEA Studio | *Python, SciPy, Three.js, WebGL, React, Arc-Length Continuation, ROM, DNV Standards* Live Demo

- Engineered an advanced computational structural mechanics platform supporting geometrically and materially nonlinear continuum regimes, multiscale homogenization, and structural optimization.
- Formulated Total and Updated Lagrangian kinematics (Green-Lagrange strains, 2nd Piola-Kirchhoff stresses) coupled with **Crisfield/Riks cylindrical arc-length continuation** to trace snap-through and snap-back buckling paths.
- Derived return-mapping J_2 plasticity with isotropic/kinematic hardening and **Lemaitre continuum damage mechanics** ($\dot{D}$), accompanied by Zienkiewicz-Zhu adaptive mesh refinement (AMR).
- Integrated **SIMP topology optimization** with MMA optimizer and manufacturing overhang constraints; computed asymptotic microstructural homogenization (FE^2) with Voigt/Reuss/Hashin-Shtrikman bounds.
- Developed snapshot **POD-Galerkin reduced-order models (ROM)** for real-time load sliders and built an interactive Three.js 3D WebGL CAD interface visualizing stress tensors, plastic strain, and ASTM E1049 rainflow fatigue life.

Relevant Coursework & Academic Highlights

Computational & Mathematical Foundations: Numerical Methods in Engineering, Finite Element Analysis, Probability & Stochastic Processes, Linear Algebra, Multivariable Calculus, Optimization Techniques, Differential Equations
Structural & Physical Mechanics: Structural Analysis, Solid Mechanics, Continuum Mechanics, Fluid Mechanics, Structural Dynamics, Mechanics of Materials, Probabilistic Risk Assessment
Research Interests: Scientific Machine Learning (Physics-Informed Deep Learning & Operator Learning), Structural Health Monitoring, Computational Fracture Mechanics, Multi-Hazard Digital Twins, and Offshore Engineering Reliability